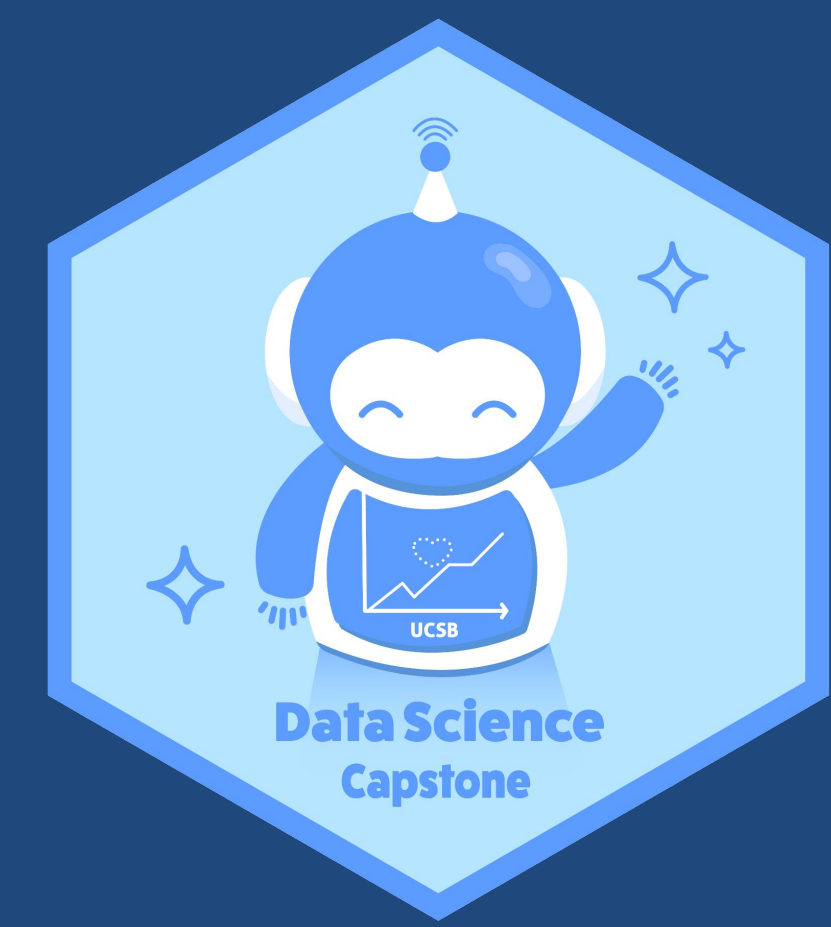


Reconstruction of a Simulated Borehole from Unstructured Particle Data

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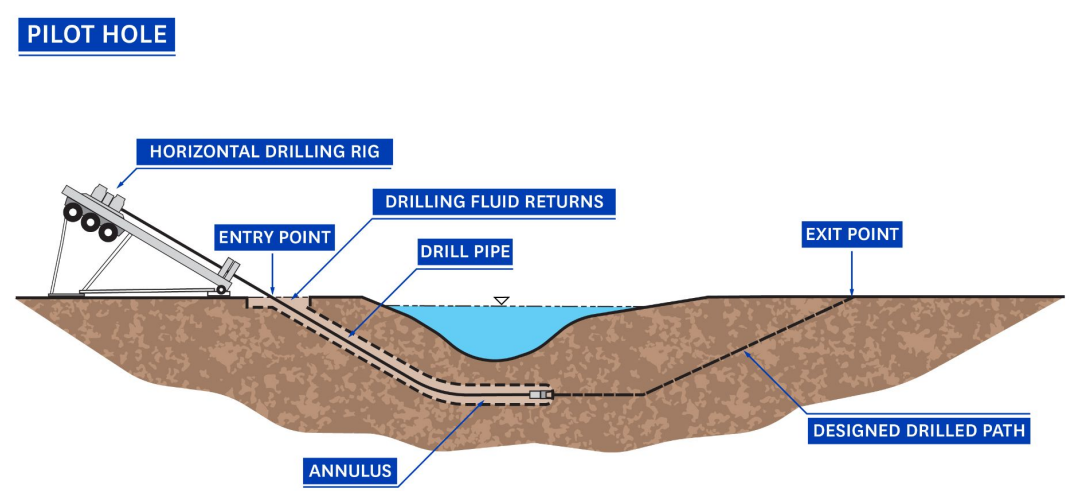
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Abstract

Geometric characterization of boreholes is a critical step for horizontal directional drilling (HDD) path-planning and reactive decision-making. This project addresses the challenge of reconstructing a smooth, continuous borehole surface from the noisy, unstructured 3D particle data generated by HDD simulations, in which the borehole exists only as a low-density void with no defined surface. We developed a slice-based pipeline that segments cross-sectional density fields, applies gradient-based candidate selection, and fits elliptical cross-sections using a RANSAC-based implementation robust to particle noise and outliers. Interpolation bridges gaps between valid slices, and aspect-ratio pre-filtering eliminates poor fits. The result is an exportable 3D mesh that reliably estimates borehole geometry and informs engineering decision-making in the field.

Introduction

Horizontal directional drilling (HDD) is widely used to install underground infrastructure without surface excavation. Physics-based simulations are used to model soil displacement during drilling, but their output is an unstructured 3D particle cloud in which the borehole exists only as a low-density void within dense surrounding soil.



- **Surface Fitting:** the borehole cross-section at each depth is modeled as an ellipse. Ellipse parameters are estimated from candidate boundary points using Principal Component Analysis (PCA). Cross-sections are stacked to form a 3D mesh.
- **RANSAC:** Random Sample Consensus is a model-fitting algorithm designed for datasets with many outliers. It repeatedly samples minimal point subsets, fits a candidate ellipse, and gives it a score calculating the points within a residual threshold. This returns the model with the highest inlier consensus.

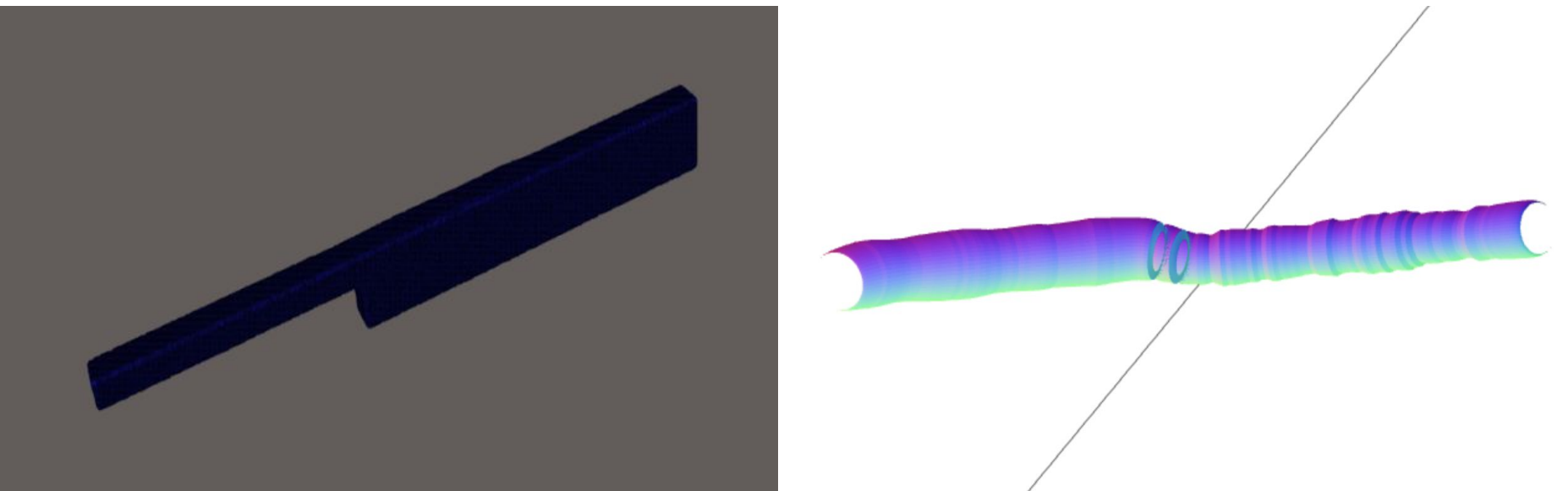


Fig. 1. High-resolution point cloud (~700k particles). Each point represents displaced soil surrounding the borehole cavity. The interior appears as a low-density void.

- **Point Cloud:** a set of 3D spatial coordinates (x, y, z) representing scanned or simulated material
- **Borehole cavity:** the hollow region created by drilling, appears as a low-density zone in the point cloud surrounded by dense soil particles

Datasets

1. **Soil Particle Point Cloud (VTK)** and a **Drill-head Reference Mesh (STL)** were generated from a granular physics simulation of horizontal directional drilling (HDD) provided by **Singularity Solutions**, in which a drill string penetrates a soil domain and displaces particles to form the borehole.
2. The simulation output was exported as raw VTK and STL files and processed locally; all data is **simulation-generated**, with no physical field measurements collected.

Why is the borehole hard to identify?

The borehole has no recorded boundary or surface and only a local drop in particle density distinguishes void from soil. The geometry must be reconstructed entirely from spatial structure in the points, rather than read from a labeled value.

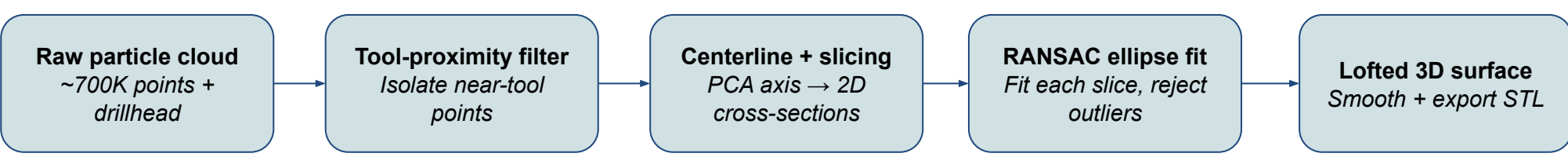
What does each data point actually contain?

Only a 3D spatial coordinate (x, y, z) per soil particle and lacks density, material, or boundary attributes. The drill-head mesh adds a separate geometric reference for locating particles along the drill path.

How large and how clean is the data?

Roughly 700,000 particle coordinates span the full simulated domain. Spacing is non-uniform and noisy, and the void boundary is ambiguous in sparsely sampled regions.

Borehole Reconstruction Pipeline



Methodology

1. **Centerline extraction:** Principal component analysis is applied to the near-tool particles to recover the dominant drill axis. Particles are projected onto this axis and binned into stations, and the per-station centroids form a centerline that is smoothed to define a stable sequence of cross-sectional slicing planes along the drill path.
2. **Slice and project:** The mirrored 3D VTK point cloud was sampled using thin slices along the x-axis, and each slice was projected into the (y, z) plane.
3. **Detect the cavity:** For each slice, a 2D density histogram was built, smoothed with a Gaussian filter, thresholded for low-density regions, and cleaned morphologically.
4. **Remove exterior regions:** Any low-density component touching the image boundary was rejected so the exterior empty region was not mistaken for the borehole cavity.
5. **Fit the cross-section:** Boundary-near points from the selected interior cavity were fit with an ellipse using PCA initialization followed by nonlinear refinement.
6. **Reconstruct the borehole:** Candidate fits were scored using residual error, inlier ratio, and slice-to-slice continuity. Short internal gaps were interpolated, and the accepted slice rings were stitched into a watertight 3D mesh.

Key Equations:

Coordinate transform	Ellipse model
$\mathbf{u} = R^T(\mathbf{p} - \mathbf{c})$	$\left(\frac{u_x}{a}\right)^2 + \left(\frac{u_y}{b}\right)^2 = 1$
Algebraic residual	Geometric residual
$\rho = \sqrt{\left(\frac{u_x}{a}\right)^2 + \left(\frac{u_y}{b}\right)^2}$	$r(\mathbf{p}) \approx \rho - 1 \cdot \left(\frac{a+b}{2}\right)^2$

Results

The slice-based reconstruction pipeline successfully inferred a continuous borehole geometry from a noisy 3D VTK point cloud containing approximately 696,791 points. After mirroring, the reconstruction used 1.39 million points of support and produced a conservative, watertight 3D mesh.

In the most recent reported run:

- 31 slice fits were accepted directly from data
- 2 short internal gaps were repaired by interpolation
- 33 total cross-sections were used in the final reconstruction
- Final mesh had 0 boundary/nonmanifold edge segments

Key quantitative results:

- Observed slice fits: 31
- Interpolated slices: 2
- Total reconstructed sections: 33
- Mean inlier ratio: 0.944
- Median adjacent-slice IoU: 0.974
- Final mesh boundary/non manifold edges: 0

Fig. 1: Representative slice showing low-density cavity selection and refined elliptical borehole fit in the (y, z) plane.

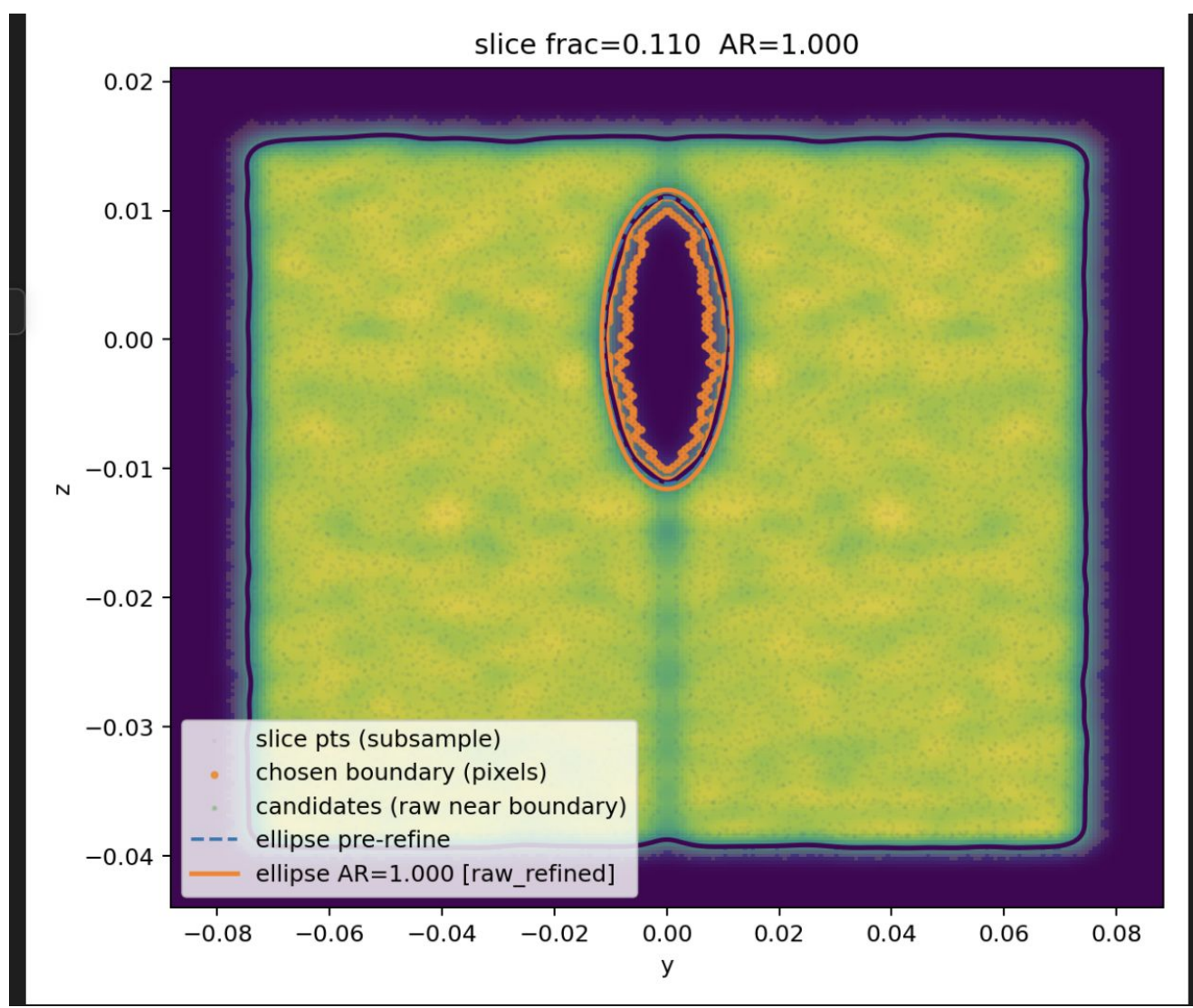


Fig.2: Borehole continuity metrics across accepted slices. Slice-to-slice overlap remained high overall, with limited center and axis drift except in weaker end regions.

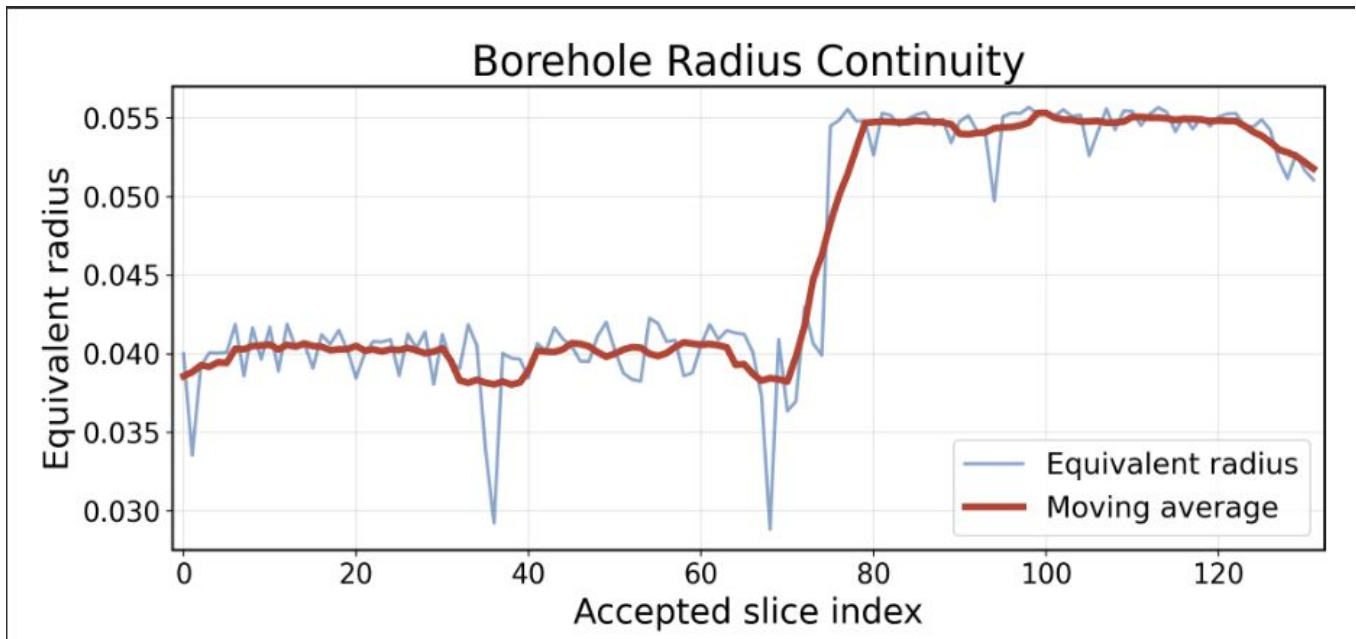
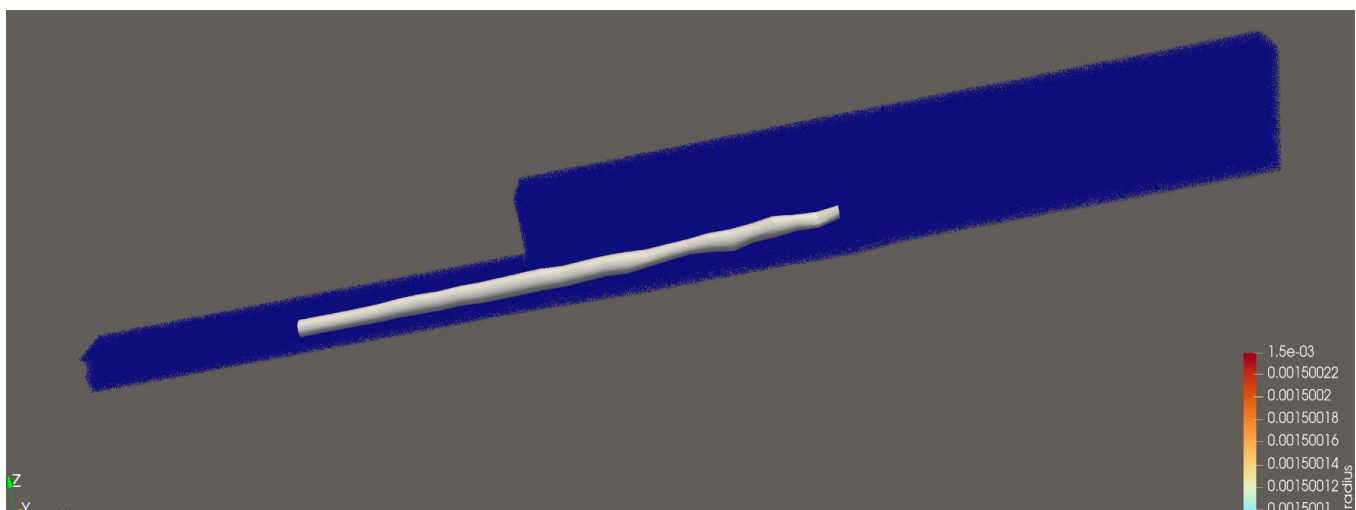


Fig. 3: Final borehole reconstruction after short-gap interpolation and strip lofting; the exported mesh was watertight.



Summary of findings

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- **Stable borehole inference:** The slice-based pipeline successfully reconstructed a continuous borehole geometry from indirect low-density structure in a noisy 3D point cloud.
- **Key segmentation improvement:** Rejecting any low-density component that touched the slice boundary prevented the exterior empty region from being mistaken for the borehole cavity.
- **More reliable fitting:** PCA initialization followed by nonlinear ellipse refinement was more stable than the earlier RANSAC-based approach, especially in noisy or weak slices.
- **Improved continuity:** Metric-driven fit selection and short-gap interpolation reduced local failures while avoiding unsupported extrapolation at the borehole ends.
- **Watertight 3D output:** The final reconstruction produced a conservative watertight mesh, with zero boundary/nonmanifold edge segments in the strongest runs.

Status Quo: The slice-based reconstruction pipeline achieved its core goal of converting an unstructured, unlabeled point cloud into a continuous, CAD-ready borehole surface. Continuity metrics and short-gap recovery showed that the reconstructed geometry was stable and physically plausible across the supported drill path. The final output was conservative and watertight, but precise agreement with true wall geometry still requires further validation.

Future Work

1. **Interactive 3D bore viewer** → turn the reconstructed borehole into a tool clients can rotate, slice open, and measure, so they can inspect drilling results without touching raw data
2. **Compare drill-heads** → run the reconstruction with different drill-head shapes to show how each one changes the size and smoothness of the hole it produces
3. **More automation** → let the pipeline tune its own settings instead of relying on manual adjustment, moving toward a one-click reconstruction
4. **Flat-terrain extension** → adapt the method to flat ground to visualize how soil compresses at the surface, a feature one customer specifically requested
5. **Other statistical methods** → explore bayesian optimization to account for noise in the simulation, producing the borehole shape along with a confidence range rather than a single best guess

References and Acknowledgments

Martinez-Otzeta, J. M., Rodríguez-Moreno, I., Mendialdua, I., & Sierra, B. (2023). RANSAC for Robotic Applications: A Survey. *Sensors*, 23(1), 327. <https://www.mdpi.com/1424-8220/23/1/327>

Computational analyses used Python with PyVista and trimesh for 3D point-cloud and mesh handling, NumPy and SciPy for spatial queries and numerical processing, and scikit-image and scikit-learn for RANSAC ellipse fitting and PCA-based centerline extraction. The reconstructed borehole surface was exported as a CAD-ready STL mesh.

Handout linked here:

